

PRANAY SONI

SENIOR COMPUTER VISION, MACHINE LEARNING & AI
ENGINEER

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PROFESSIONAL PROFILE

Seasoned computer vision senior engineer with experience designing, developing, and deploying production-grade vision systems, taking ideas from research and algorithm development through high-performance prototypes and reliable product delivery. Deep expertise in image processing, detection, segmentation, anomaly detection, multi-camera vision, and real-time perception, with hands-on experience across classical CV, CNNs, ViTs, YOLO, SAM2, DINO vision foundation models. Strong C/C++ and Python background, with experience optimizing AI pipelines for real-world hardware constraints using OpenVINO, TensorRT, and CUDA. Co-founded and scaled a machine vision company, leading cross-functional R&D teams and delivering 10+ production systems used across highly regulated pharmaceutical and manufacturing environments.

CORE COMPETENCIES

Technical: Computer Vision & Perception: Classical & deep-learning computer vision (CNNs, ViTs, YOLO, SAM2, U-Net), Image processing, Object detection & segmentation, Anomaly detection, Algorithm design • **AI/ML:** Deep learning, Transformer architectures, Supervised & unsupervised learning (classification, regression, clustering), Predictive-AI workflows (data preprocessing, feature engineering & selection, model training, tuning & evaluation), Ensemble methods & model validation (cross-validation, ROC/AUC, bias-variance) • **Edge & Deployment:** Model quantization, Edge computing, Real-time AI inference, C/C++ & Python • **Generative AI:** LLMs, RAG pipelines using knowledge graphs and vector databases (Neo4j), Generative AI & prompt engineering

Leadership & Soft Skills: Cross-functional R&D, Academic & industry collaborations, Technical documentation, Stakeholder communication, Customer-centric product development, Team collaboration, Mentoring

WORK EXPERIENCE

SPAN INSPECTION SYSTEMS PVT LTD, GANDHINAGAR, INDIA, SEPTEMBER 2015 TO PRESENT (*FRACTIONAL / ADVISORY*)

R&D DIRECTOR & CO-FOUNDER

Co-founded and helped scale the company across a decade; now stepped back from day-to-day operations while retaining a minority stake during a phased exit.

- Pioneered the industry's first auto-teaching capability in inspection products, reducing recipe creation time by 80% through advanced image processing techniques and AI.
- Delivered around ten production vision systems from concept to deployment for Fortune 500 clients, grounding each in direct user research to drive market adoption and address real pain points.
- Architected and implemented over one million lines of production-quality C++ and Python algorithmic code powering reliable machine vision systems deployed globally.
- Developed and Architected end to end self-evolving GraphRAG pipeline which can answer customer technical queries based on the knowledge base.
- Designed and deployed advanced computer vision pipelines for segmentation, object localization, and anomaly detection, optimized for multi-core CPU and GPU edge devices to achieve sub-100ms processing latencies.
- Led and mentored a cross-functional team of researchers and engineers, consistently meeting project deadlines while raising the technical standard and fostering a collaborative, high-performance culture.
- Built research partnerships with academic institutions that strengthened the R&D pipeline, directly enabling three filed patents and a peer-reviewed publication in an Elsevier journal.
- Developed breakthrough mechanical and opto-mechanical design innovations that enabled machines to outperform competitors by 30% on key metrics, applying detailed observational analysis and nature-inspired principles.

JEKSON VISION, AHMEDABAD, INDIA, APRIL 2006 TO SEPTEMBER 2015

R&D LEAD

- Led development of proprietary image processing libraries tailored for resource-limited environments, eliminating third-party dependencies and optimizing performance on custom hardware platforms.
- Architected complete image processing pipelines spanning pre-processing (noise filtering, histogram equalization) to high-level semantic segmentation and pattern recognition, enhancing reliability and throughput of industrial inspection systems.
- Led an R&D team to deliver multiple market-leading innovations that generated \$4M in new revenue by establishing clear research priorities and fostering technical excellence.
- Developed first-of-its-kind tablet-inspection system on bulk lines through three years of rigorous R&D in software and opto-mechanical systems, driving a 50% boost in company turnover.
- Oversaw the full product lifecycle for multiple systems — from requirements collection through development, deployment, and customer support — ensuring end-to-end quality and client satisfaction.

EDUCATION

Master of Management in AI | Queen's University Smith School of Business, Toronto, ON | 2026 – 2027

Bachelor of Engineering (IT) | Gujarat University, India | 2006

Certification: Advanced Certification Program in Generative AI and Prompt Engineering, IISc Bangalore, India (2024)

PATENTS/PUBLICATIONS

Patents (Approved by Intellectual Property of India): Method and system of automatic inspection of blister pack (595397)

Patents (Provisional): Track & Trace machine with improved changeover (202621010729) • Print & Check machine with improved changeover (202621007538) •

Publications: “A novel deep learning pipeline for coronary artery analysis in X-ray angiography,” Engineering Applications of Artificial Intelligence (*Elsevier*, 2025). • “Transforming dataset creation: leveraging generative AI for machine vision applications in data-limited domains,” International Journal of Advances in Electronics and Computer Science.

Awards & Features: Vision Systems Design 2017 Innovators Award for automatic teaching in blister inspection systems. Featured in [Vision Systems Design magazine article](#) on vision system for blister pack inspection.

HACKATHONS

Hack the North 2026 First prize Winner in sponsors category from Embedded Systems with AI by QNX

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python (production-quality, performance-critical); scikit-learn, LangGraph, OpenAI API, MCP, pandas, NumPy, PyTorch, TensorFlow, OpenCV, ONNX Runtime, OpenVINO, TensorRT, LangChain, FastAPI, Optuna, Neo4j, Ollama.

ADDITIONAL INFORMATION

Languages: English, Hindi, Gujarati